

LEISHMANIA

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LEISHMANIA

- Causative agent of Leishmaniasis
- Obligate intracellular protozoa of the genus *Leishmania*
- In the human host, *Leishmania* are intracellular parasites that infect the mononuclear phagocytes

LEISHMANIA

- Leishmaniasis can easily be classified clinically as
 - Visceral leishmaniasis
 - Cutaneous leishmaniasis
 - Mucocutaneous leishmaniasis
 - Diffuse cutaneous leishmaniasis
- These different forms of the disease are caused by the different species of *Leishmania*

LEISHMANIA

- **Cutaneous leishmaniasis(CL)**

- *L. tropica*
- *L. major*
- *L. aethiopica*
- *L. panamensis*
- *L. guyanensis*
- *L. peruviana*

- **Visceral leishmaniasis(VL)**

- *L. donovani*
- *L. infantum*
- *L. Chagasi*

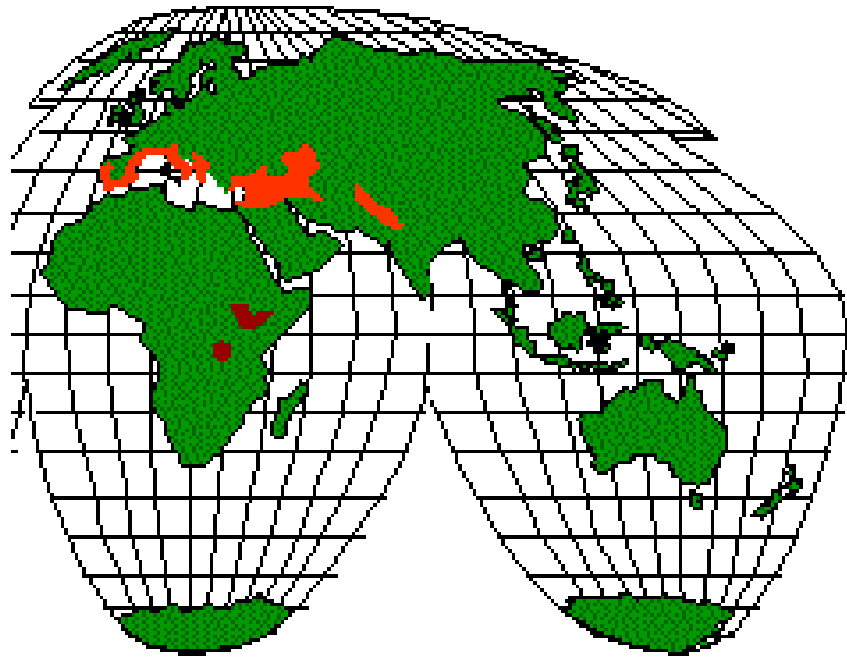
- **Mucocutaneous leishmaniasis(MCL)**

- *L. panamensis*
- *L. guyanensis*
- *L. Brazilliensis*

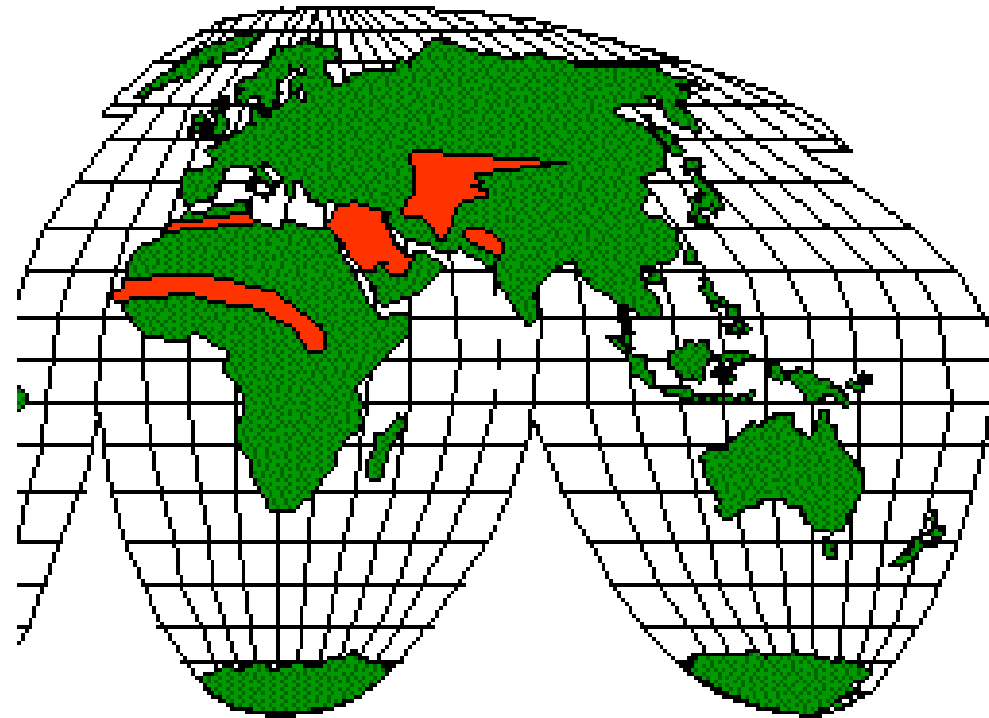
- **Diffuse cutaneous leishmaniasis(DCL)**

- *L. amzonensis*
- *L. aethiopica*

Epidemiology



■ *L. aethiopica*
■ *L. tropica*



■ *L. major*

Epidemiology

- Geographical distribution is limited by:
 - The distribution of the sandfly,
 - Its tendency to take blood from humans or animals only, and
 - Its capacity to support the internal development of specific species of *leishmania*

Leishmaniasis incidence increases; mainly

- Man-made environmental changes
- Poverty and malnutrition
- Movement of susceptible populations into endemic areas

In Ethiopia

- Four species of *Leishmania*,
 - *L. aethiopica*,
 - *L. major*
 - *L. tropica*
 - *L. donovani*

In Ethiopia

- **Cutaneous leishmaniasis**

- Endemic at altitudes between 1400 and 2700 m
- Prevalence rates of 5.5 – 40%
- In **Shewa**, **Wello** and **G.Gofa**
- Reservoir host: Rock hyrax (*Procavia habessinica*) and tree (*Heterhyrax brucei*) hyraxes serving as for *L. aethiopica*

Visceral leishmaniasis (VL)

Occurs mainly in arid and semiarid lowlands below 1,300 m altitude

Important endemic foci include

- **Gelana focus at lake abaya,**
- **Aba- Roba focus in Knoso Wereda**
- **Omo river** plains
- **Metema and Humera** plains in north western Ethiopia

Transmission and life cycle

- Common mode of transmission.
 - Bite of **sandfly**
- Uncommon modes of transmission:
 - Congenital transmission,
 - Blood transfusion,
 - Rarely, inoculation of cultures

Life cycle: Two Life-cycle stages

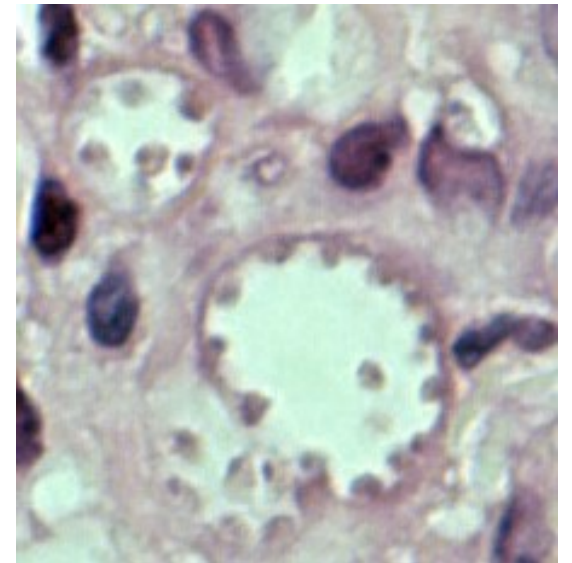
Promastigote

- Elongated, with flagella
(10-20 μm long)
- Occur extracellularly in the
-insect midgut & in artificial culture
- Motile

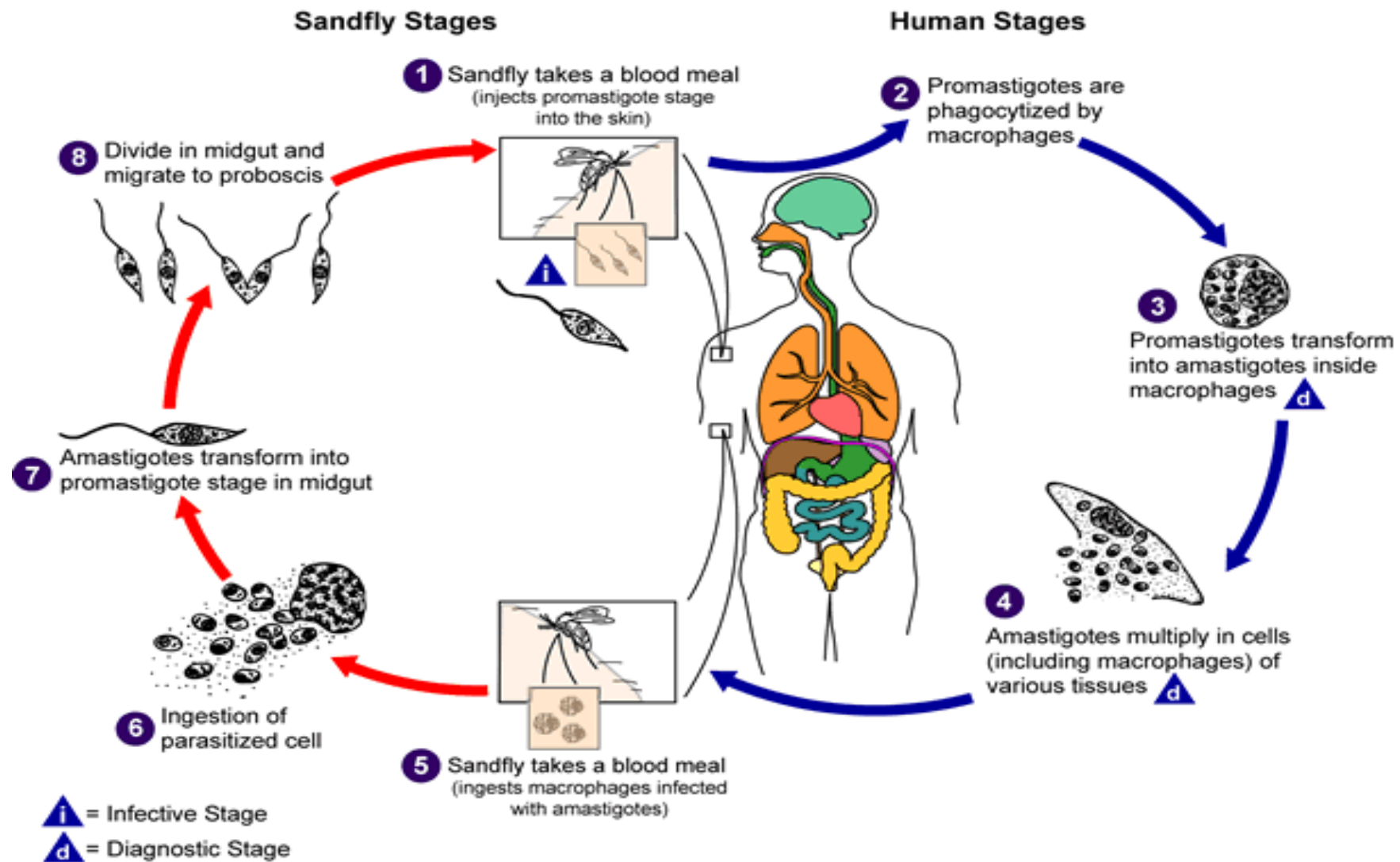


Amastigote

- Round (3-7 μm diameter)
- Occurs intracellularly, during
mammalian stage
- Non-motile



- Promastigotes are phagocytized by macrophages & transform into intracellular amastigotes form
 - ✓ **In VL** the amastigotes are carried through blood circulation , then invade and multiply in the macrophages of **spleen, liver, Bone marrow, lymph glands** , etc.
 - ✓ **In CL and MCL** – the amasigote multiply in skin macrophages (histocytes)
- Sandflies become infected during blood meals when they ingest macrophages infected with amastigotes
- The host cell break down and releasing the amasigotes which is then transform to promastigotes
- Multiply , fill the lumen of the gut and migrate to the proboscis



- Entrance into the host and establishment of infection by leishmaniasis is enhanced by saliva from the vector
- Two substances were involved
 - ✓ **maxadilan**, or maximum dilation molecule: keeps the capillary bed open
 - ✓ **SIP** or salivary immunosuppressive protein : restrains the immune system's early efforts to eliminate the parasites

- phagosome + lysosome to form a **phagolysosome**
 - ✓ As the **promastigotes** transform into **amastigotes**, which produce compounds that counter lysosomal enzymes
 - ✓ The **gp 63** molecule inactivates proteolytic enzymes
 - ✓ **LPG** protects against other enzymes

Cutaneous Leishmaniasis

Causative agents

Leishmania tropica

Leishmania major

Leishmania aethiopica

Leishmania mexicana

Leishmania peruriana

Leishmania panamensis

Leishmania guyanensis

Cutaneous Leishmaniasis



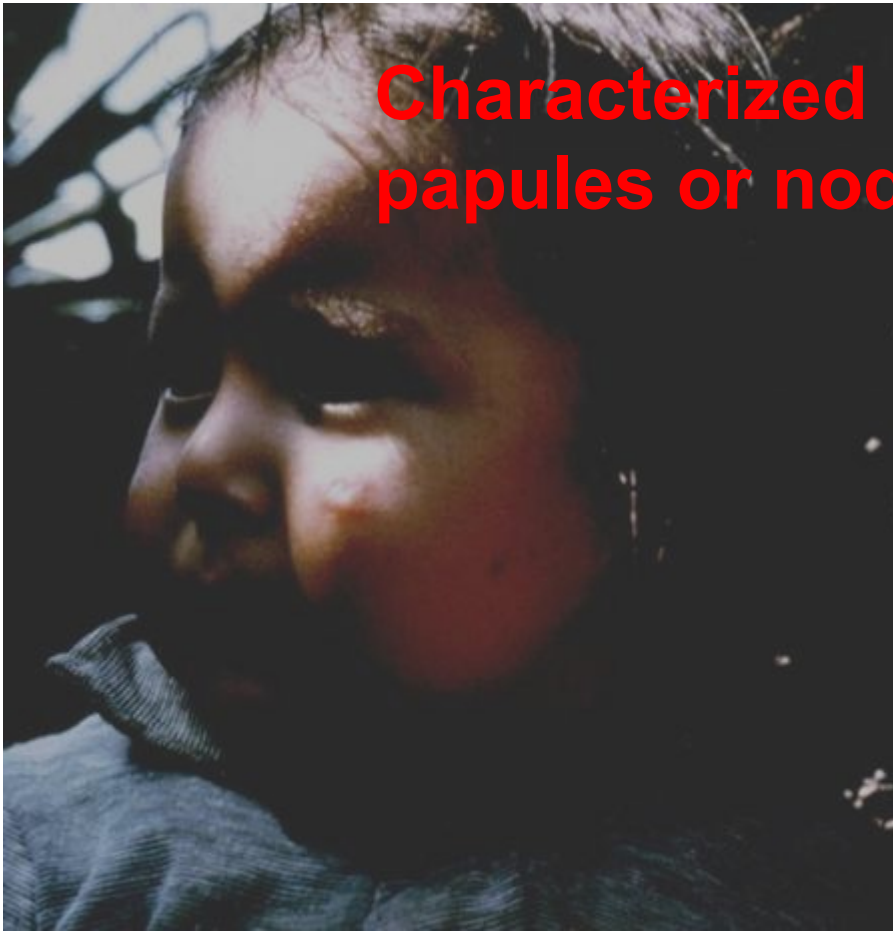
a) Acute

- Incubation period: **2 weeks to several months**
- Initially, the lesion is a small, red papule up to 2 cm in diameter
- change in size and appearance over time



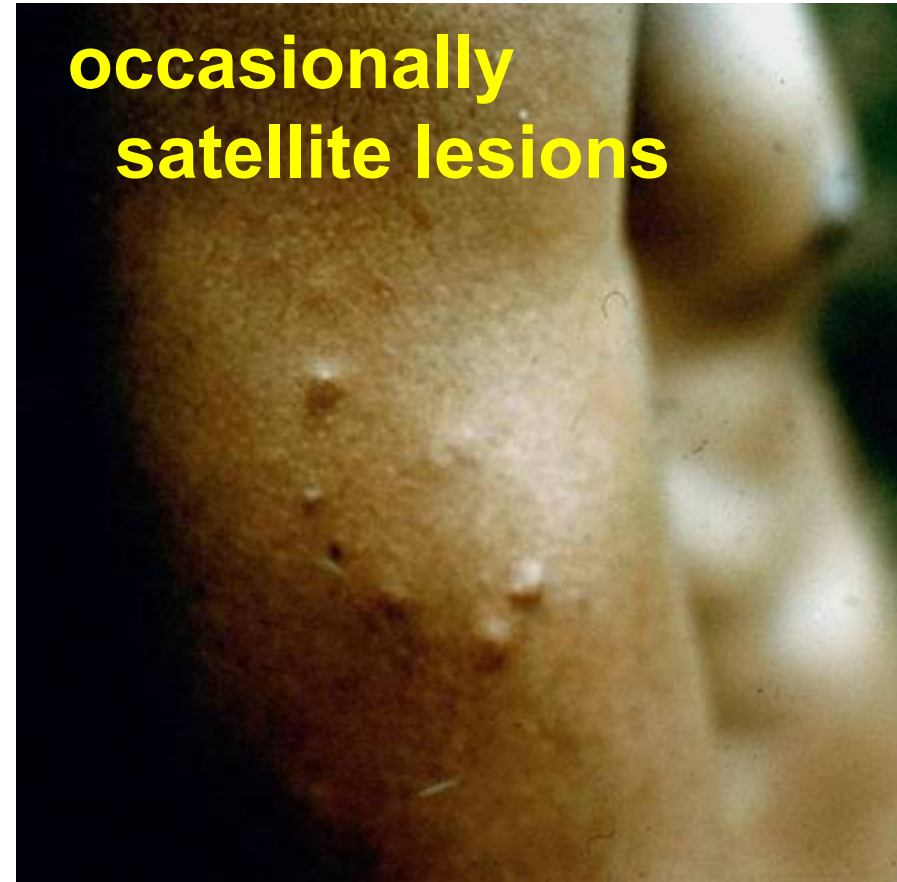
b) Chronic

- Chronic ulcerated, papular, or nodular lesion
- lesion is painless, non-tender, non-pruritic and usually clean



**Characterized by one or more sores,
papules or nodules**

**ulcerated, papular, or
nodular lesions**



**occasionally
satellite lesions**



Cutaneous Leishmaniasis

- chronic ulcerated, popular, or nodular lesion
- lesion is painless, non-tender, non-pruritic and usually clean



Cutaneous Leishmaniasis

- self-healing, months to years
- Sores can leave significant scars and be disfiguring if they occur on the face



Cutaneous Leishmaniasis

- Often described as looking somewhat like a volcano with a raised edge and central crater
- occasionally palpable lymph nodes



The clinical forms of CL vary according to

- The species of parasite
- Region and
- Response of patient



Old World CL

L. Tropica (SW Asia, N.Africa

- Anthroponotic or dog reservoir
- dry urban oriental sore
- 'dry painless lesion'
- 25-70mm diameter

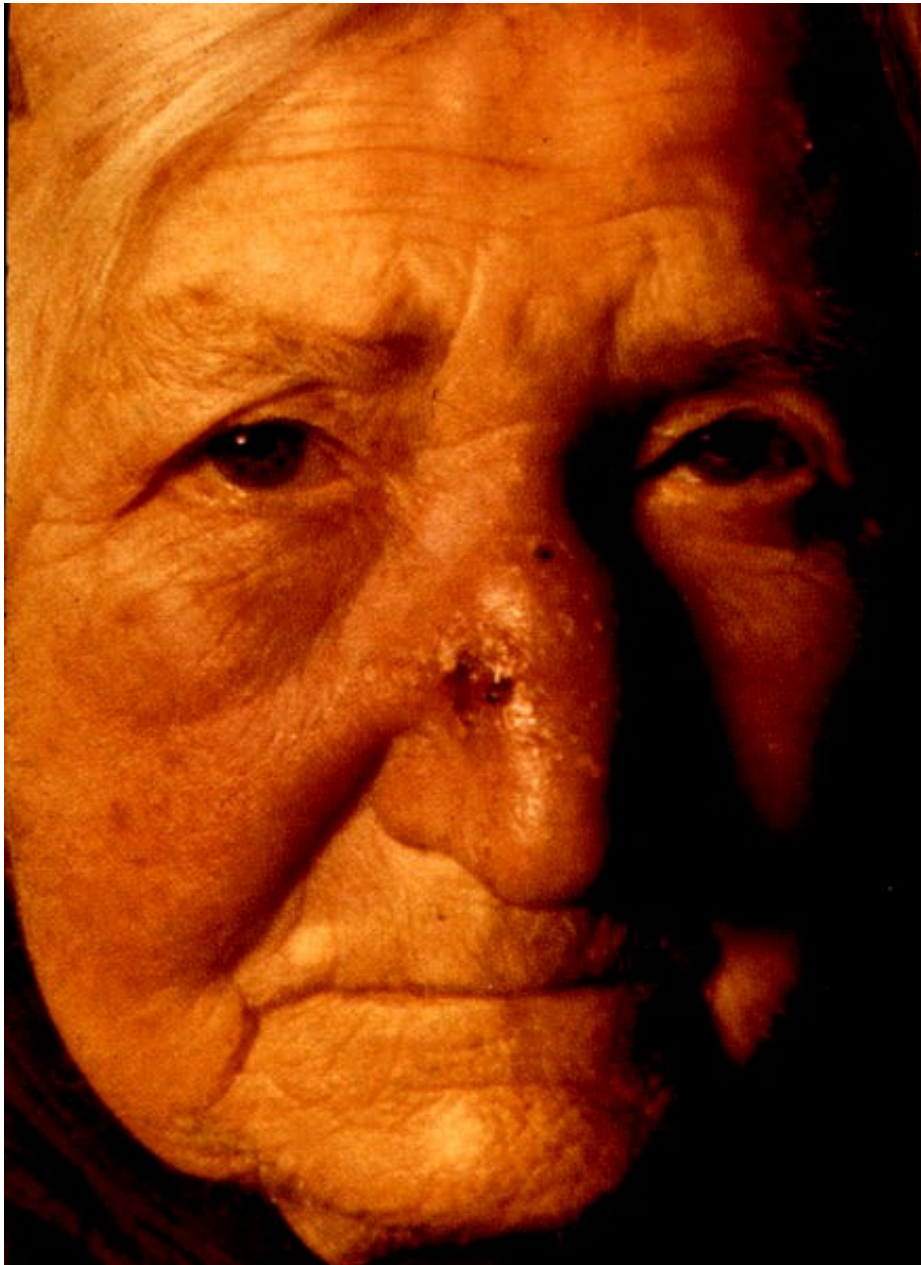
Old World CL



L. Major (Wet oriental sore)

- Central Asia, middle East, Africa
- Rural (rodent reservoir)
 - Wet lesion
 - Early papules is inflamed(5-10mm)
 - Develop to large uneven ulcer
 - Self-healing (3-6mths)
 - Protect against reinfection & also with *L.tropica*

Ulcers are moist or open with seropurulent exudate



Old World CL

L. infantum

- Mediterranean, Europe
- dermatrophic strains recently recognized

- *L. aethiopica*

- highlands of Kenya and Ethiopia
- Similar to oriental sore
- Self-heal 1-3 yrs
- Can cause DCL

Diffuse Cutaneous Leishmaniasis (*L. aethiopica*)



- Lesion develop over large areas of the body
- Scaly, not ulcerated, nodules
- Chronic and painless
- Numerous parasites in lesions
- Seldom heal despite treatment



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- Chronic and painless
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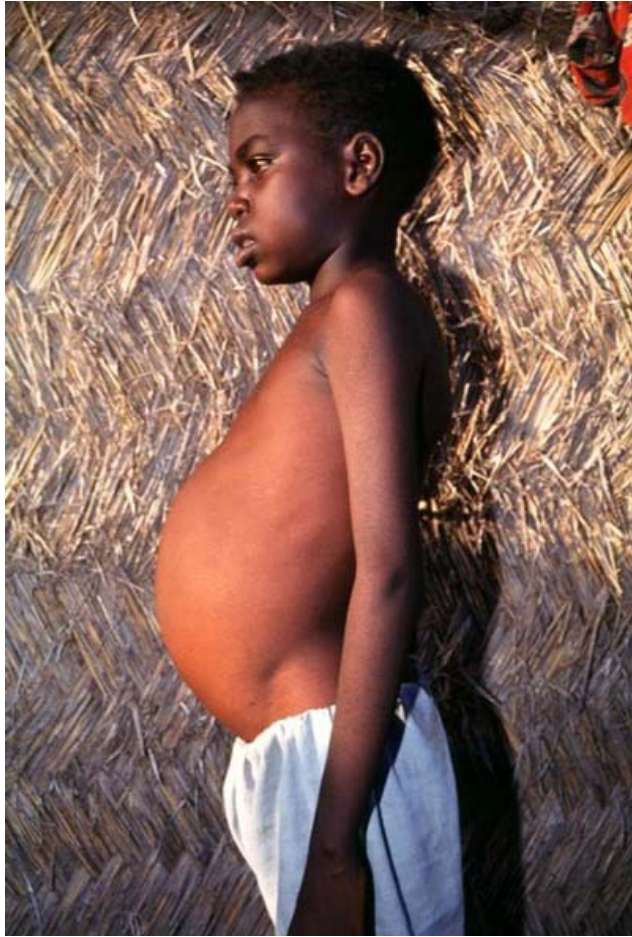
Visceral Leishmaniasis

- Caused by the *Leishmania donovani* complex,
 - ✓ *L. donovani*, *L. infantum*, *L. chagasi*
- Reticuloendothelial system affected
 - Spleen, liver, bone marrow, lymph nodes
- Progressive disease
 - 75-95% mortality if untreated
 - Death generally within 2 years
- Death usually occurs- **severe secondary bacterial infections** in advanced disease
- Pneumonia is the common complication

Clinical Presentation



- Incubation period:
 - Generally 2-6 months but can range 10 days to years
- Fever, malaise, weakness
- Wasting despite good appetite
- Spleno- and hepatomegaly, enlarged lymph nodes
- Depressed hematopoiesis
 - Severe anemia, leucopenia
 - Thrombopenia → petechial hemorrhages in mucosa



- The boy exhibits splenomegaly, distended abdomen and severe muscle wasting
 - from visceral leishmaniasis



- A 12-year-old boy suffering from visceral leishmaniasis. The boy exhibits splenomegaly and severe muscle wasting

- Jaundiced hands of a visceral leishmaniasis patient



Leishmaniasis and HIV Infection

- Coexistence of leishmaniasis with HIV infection is a serious concern
- Leishmaniasis is spreading in several areas of the world because of the rapidly spreading epidemic AIDS
- The immune deficiency has lead to increased susceptibility to infections, including leishmaniasis

- Co-infection with HIV has lead to the spread of leishmaniasis, typically a rural disease, into urban areas
- In patients infected with HIV, leishmaniasis accelerates the onset of AIDS
 - By cumulative immunosuppression and
 - By stimulating the replication of the virus
- It may also change asymptomatic *Leishmania* infections into symptomatic infections

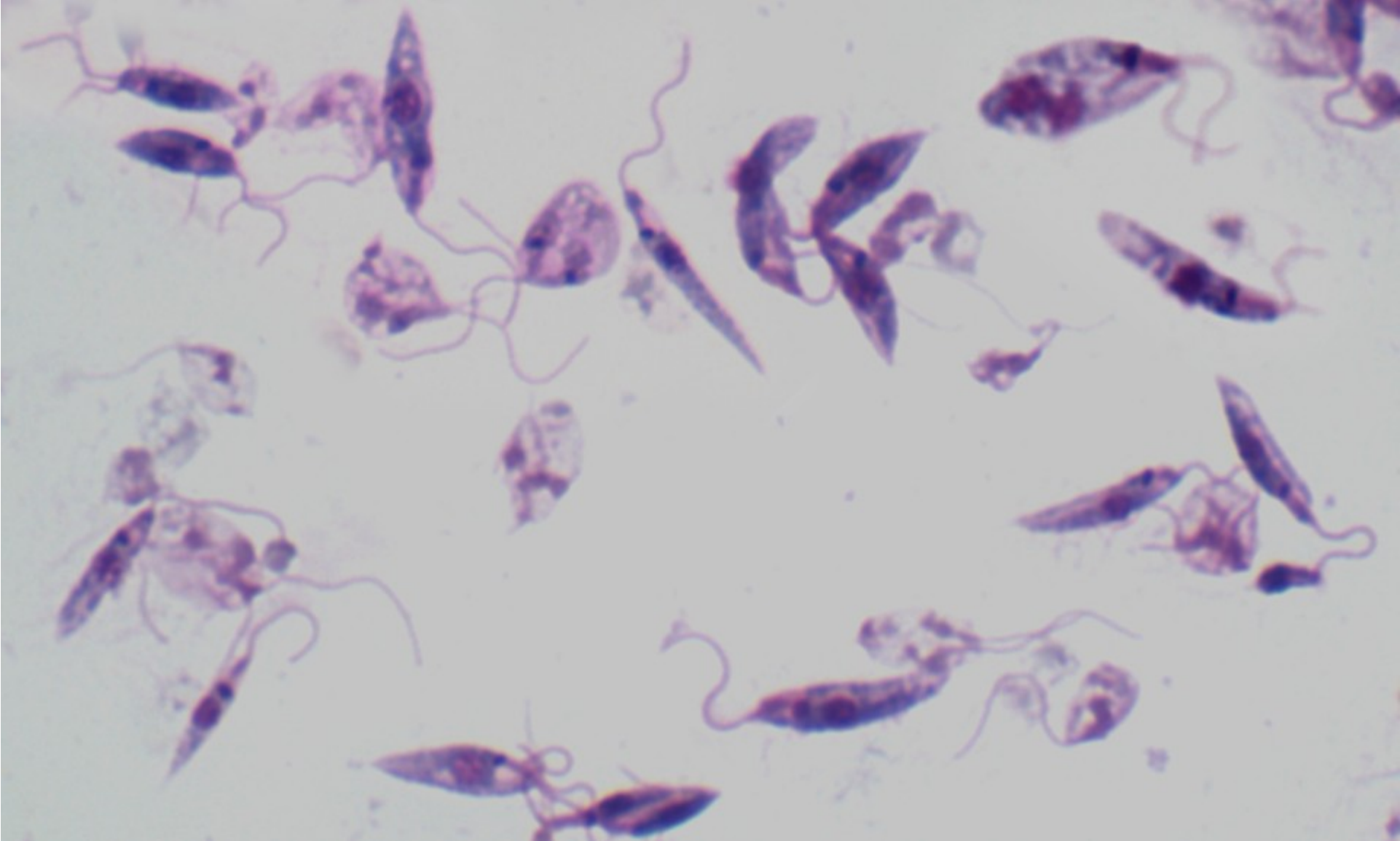
Diagnosis of CL, MCL, DCL

- suspected because of:
 - geographical presence of parasite
 - history of sandfly bite
 - + skin lesion:
 - chronic, painless, 'clean' ulcer
 - nasopharyngeal lesions
 - nodular lesions

1. Demonstration of parasite
amastigotes (scrapings, biopsy, aspirates)
2. culture from ulcer material
- 3.** Leishmainin test
4. serology?



aspiration and culture



promastigotes following in vitro culture

Treatment

- Sodium stibogluconate (Pentostam)
- Pentamidine isethionate
- amphotericin B
- cryotherapy and thermotherapy

prevent and control

1. Early detection by serological diagnosis (VL) and treatment of infected persons
1. Personal protection from sandfly bites by:
 - Using insect repellents
 - Avoiding endemic areas especially at times when sandflies are most active
 - Use of pyrethroid impregnated bed nets and curtains
3. Vector control by the use of light traps, sticky paper traps, or residual insecticide spraying of houses